May 2024
IB MAAHL P1 TZ1 Student C.

1. mean $= 3$

(a) $p + 2q + 3(4) + 4(2) + 6(3) = 3$.

$p+2q+38 = 3$

$\Rightarrow p+2q = -35$ — (1)

$p+q+4+2+3 = 16$

$p+q = 7$ —(2)

(1) − (2):

$\Rightarrow q = -42, \therefore p = 49$

(b) Mean final score $= 10(3)$

$= 30.$

2. $\log_{10} a = \frac{1}{3}$, $a > 0$

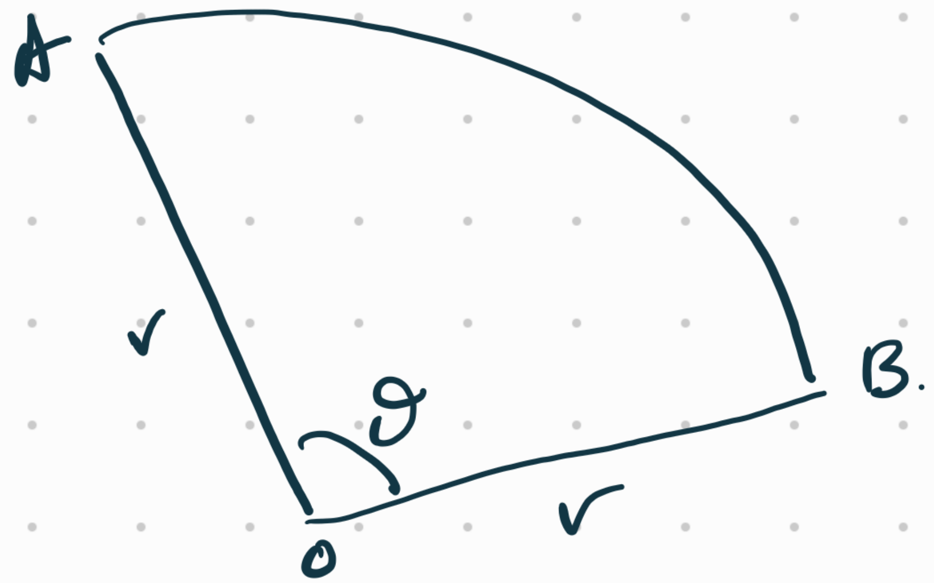
$$\log_{10}\left(\frac{1}{a}\right) = \frac{\log_{10} 1}{\log_{10} a} = \frac{0}{1/3}$$

$$= 0$$

(b) $$\log_{1000} a = \frac{\log_{10} a}{\log_{10} 1000}$$

$$= 0$$

3.



(a) perimeter $= r\theta = 10$

Area $= \frac{1}{2}r^2\theta = 6.25$

$\Rightarrow r^2\theta = 12.5$

$\Rightarrow \theta = \frac{12.5}{r^2}$ — (2)

sub. (2) into (1).

$r\left(\frac{12.5}{r^2}\right) = 10$

$12.5r = 10$.

(b) $4r^2 - 20r + 25 = 0$

$$(2r-5)(2r-5) = 0$$

$$r = \frac{5}{2}.$$

$$\theta = \frac{12.5}{r^2} = \frac{12.5}{\left(\frac{5}{2}\right)^2}$$

$$= \frac{12.5}{\frac{25}{4}}.$$

4. $f(x) = \cos x$, $g(x) = \sin 2x$

$0 \leq x \leq \pi$.

(a) $\cos x = \sin 2x$.

$\Rightarrow \cos x = 2\sin x \cos x$

$1 - 2\sin x = 0$

or $\sin x = \frac{1}{2}$.

$x = \frac{\pi}{6}$ or $x =$

x-coordinate of $A = \frac{\pi}{6}$

x-coordinate of $C =$

(b) Area of R

$$= \int_{\pi/2}^{\pi} (\cos x - \sin 2x)\, dx$$

$$= \left[\sin x - \frac{\cos 2x}{2} \right]_{\pi/2}^{\pi}$$

$$= \left[\sin\pi - \frac{\cos 2\pi}{2}\right] - \left[\sin\frac{\pi}{2} - \frac{\cos\pi}{2}\right]$$

$$= -\frac{1}{2} - \left(1 + \frac{1}{2}\right)$$

$$= 2 \text{ units}^2$$

5. G.P. $u_1 = 1$, $r = 10$.

(a) $S_n = \frac{1(10^n - 1)}{10 - 1} = \frac{10^n - 1}{9}$

$a = 10$, $b = 9$

(b) $S_1 + S_2 + S_3 + \dots + S_n$

$= \frac{10^1 - 1}{9} + \frac{10^2 - 1}{9} + \frac{10^3 - 1}{9} + \dots + \frac{10^n - 1}{9}$

$= \frac{1}{9}\left[10^1 + 10^2 + 10^3 + \dots \quad -1 - 1 - 1 - \dots\right]$

$= \frac{1}{9}\left[10 + 100 + 1000 + \dots + -n\right]$

6. $f(x) = \sqrt{x \sin(x^2)}, \quad 0 \leq x \leq \sqrt{\pi}.$

Volume of revolution about x-axis

$$= \pi \int_0^{\sqrt{\pi}/2} x \sin(x^2)\, dx$$

$$= \pi \left[-\cos x^2 \right]_0^{\sqrt{\pi}/2}$$

$$= \pi \left[-\cos \frac{\pi}{4} + \cos 0 \right]$$

$$= -\pi \left(\frac{\sqrt{2}}{2} \right)$$

7. Prove by Mathematical Induction

$$\sum_{r=1}^{n} {}^{r}C_1 = {}^{n+1}C_2 .. , \quad n \in \mathbb{Z}^+.$$

$n=1$ $\quad LHS = {}^{1}C_1 = 1 = RHS$

True for $n=1$.

let $n=k$

$$\sum_{r=1}^{k} {}^{r}C_1 = {}^{k+1}C_2 .$$

let $n=k+1$

$$\sum_{r=1}^{k+1} {}^{r}C_1 = \sum_{r=1}^{k} {}^{r}C_1 + {}^{k+1}C_1$$

$$= {}^{k+1}C_2 + {}^{k+1}C_1$$

$$= \frac{(k+1)!}{2!\,(k+1-2)!} + \frac{(k+1)!}{1!\,(k+1-1)!}$$

$$= \frac{(k+1)!}{2!\,(k-1)!} + \frac{(k+1)!}{k!} = {}^{k+2}C_2$$

Since it is true for $n=1$, and true for $n=k \Rightarrow n=k+1$ is true, by mathematical induction,

$$\sum_{r=1}^{n} {}^{r}C_{1} = {}^{n+1}C_{2}, \quad n \in \mathbb{Z}^{+}$$

8. (a) Maclaurin series:

(i) $\sin(x^2) = x^2 - \frac{(x^2)^3}{3!}$

$= x^2 - \frac{x^6}{6}$

(ii) $\sin^2(x^2) = \left(x^2 - \frac{x^6}{6} + \dots\right)\left(x^2 - \frac{x^6}{6} + \dots\right)$

$= x^4 - \frac{x^8}{6} - \frac{x^8}{6}$

$= x^4 - \frac{x^8}{3}$

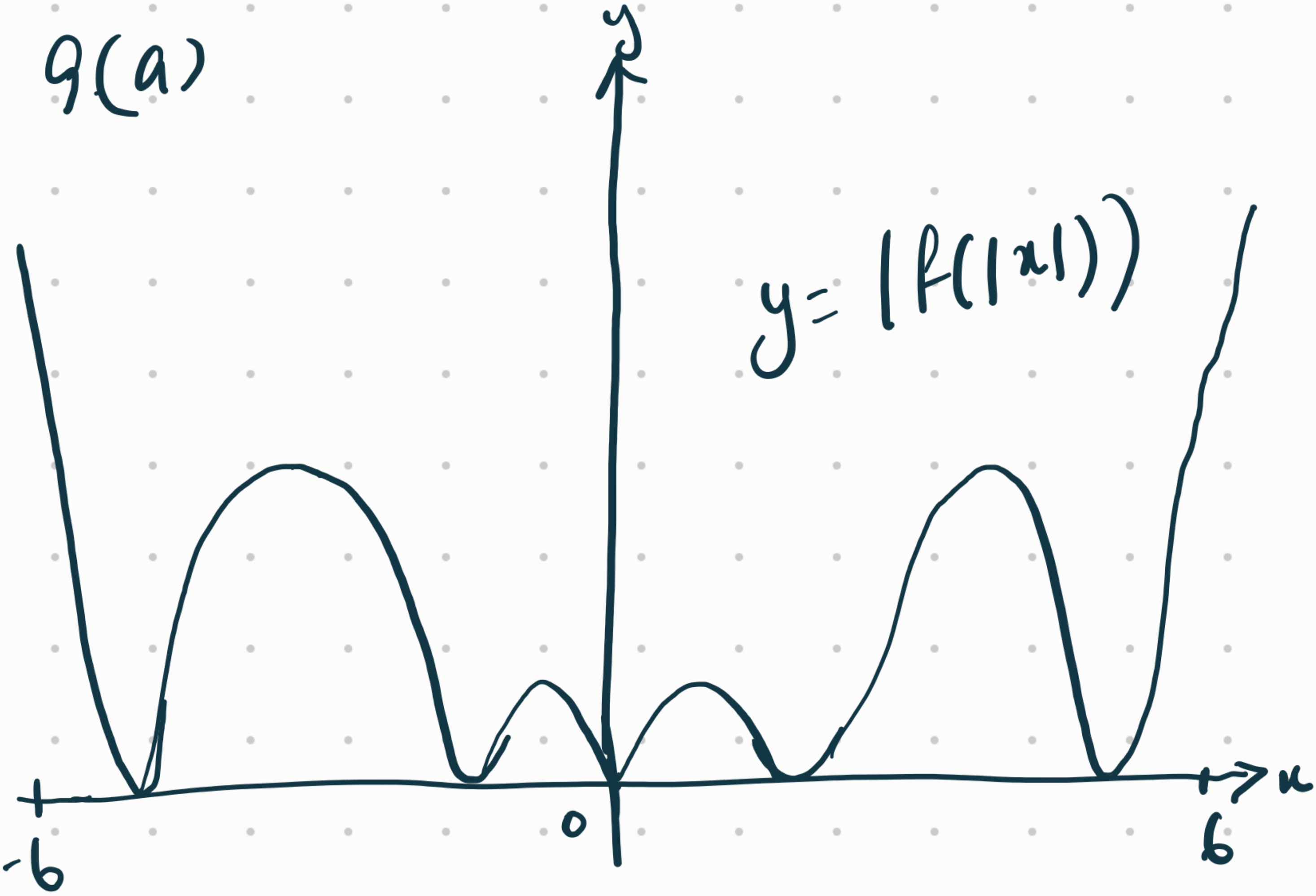
(b) $4x \sin(x^2) \cos(x^2)$

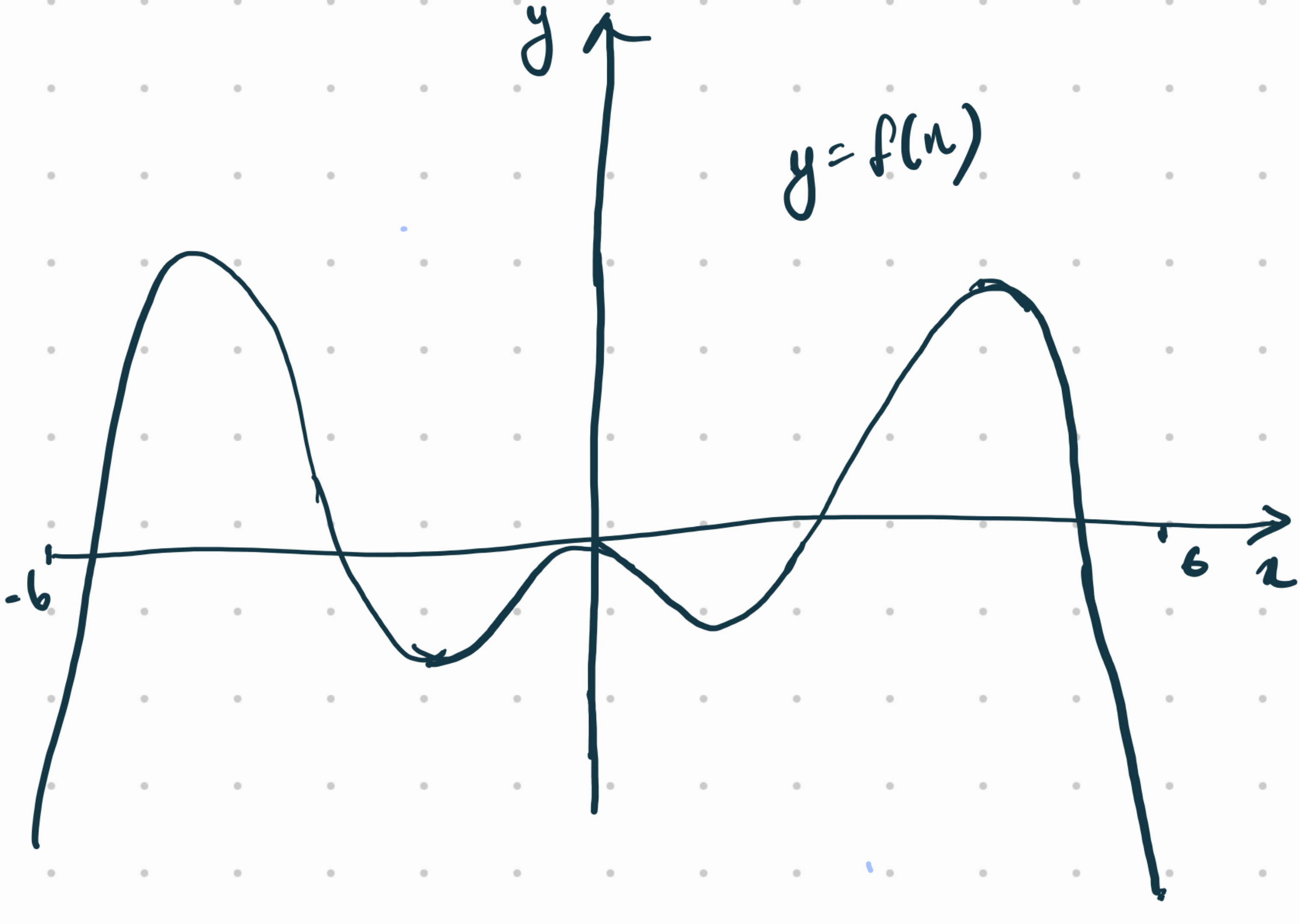
$= 4x\left(x^2 - \frac{x^6}{6}\right)\left(1 - \frac{x^2}{2!}\right)$

$= 4x\left(x^2 - \frac{x^6}{6} - \frac{x^4}{2} - \frac{x^8}{12}\right)$

$= 4x^3 - 2x^5$

9(a)





(c) $\int_0^4 f(|x|)\,dx = 1.6$

(i) $\int_{-4}^{0} f(x)\,dx = 1.6$

(ii) $\int_{-4}^{4} f(|x|) + f(x)\,dx$

$= \int_{-4}^{4} f(|x|)\,dx + \int_{-4}^{4} f(x)\,dx$

$= 3.2 + 3.2$

$= 6.4$

10. $f(x) = \dfrac{4x+2}{x-2}, \; x \neq 2.$

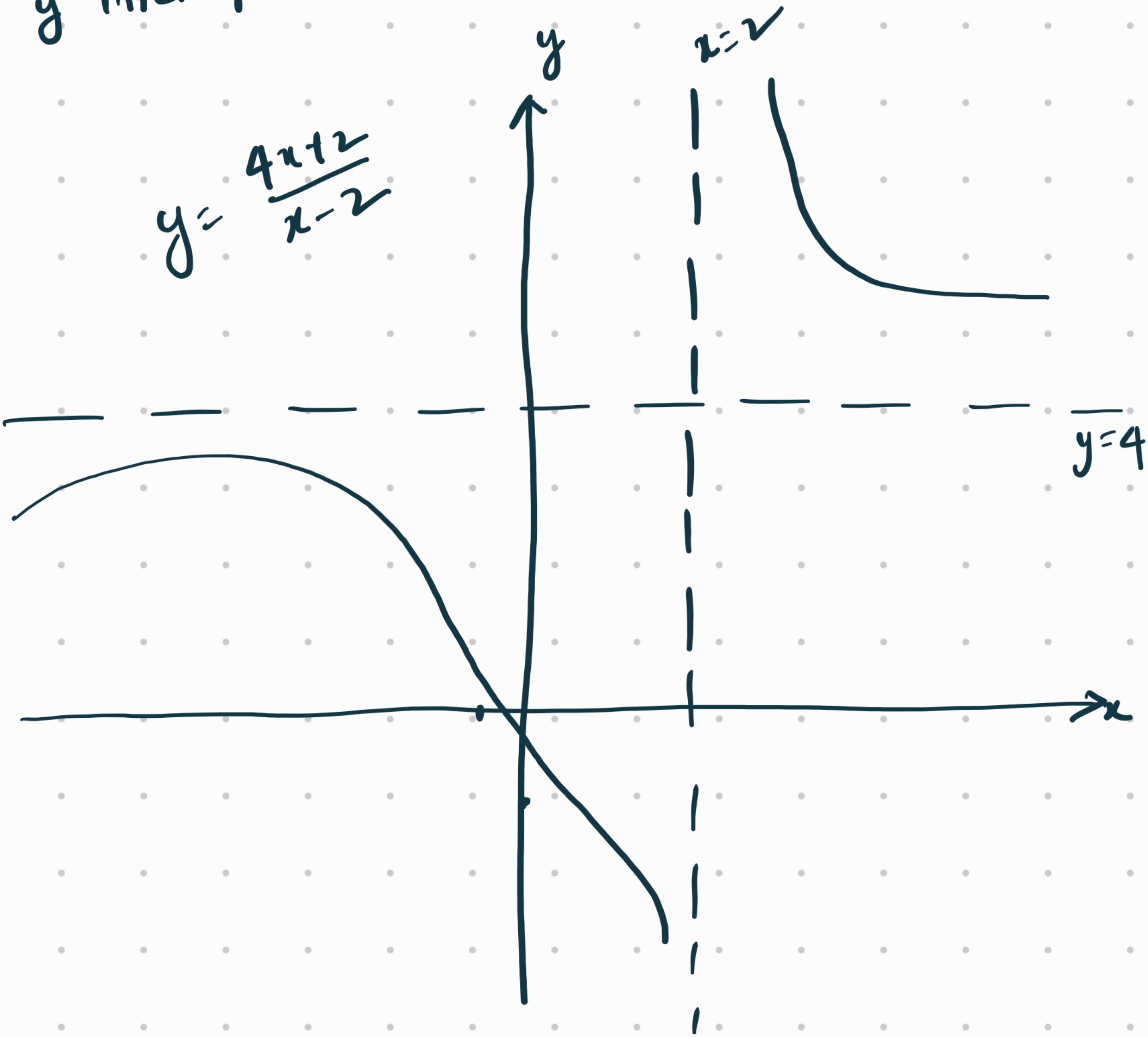
(a) $= 4 + \dfrac{10}{x-2}$

vertical asymptote: $x = 2$.

Horizontal asymptote: $y = 4$.

x intercept: when $y = 0$, $x = -\frac{1}{2}$

y intercept: when $x = 0$, $y = -1$

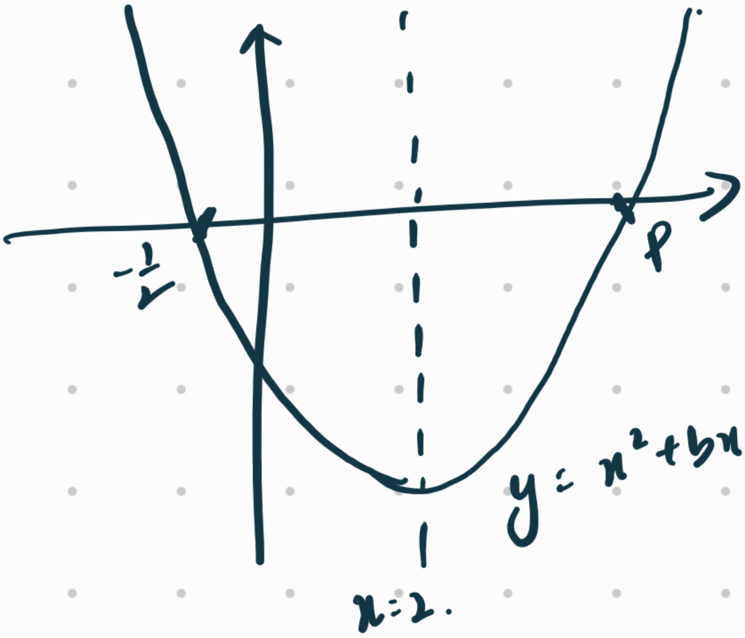


(b) Range of f.

$y \neq 4$.

(c) $g(x) = x^2 + bx + C$

Axis of symmetry : $x = 2$.



$\frac{p - \frac{1}{2}}{2} = 2 \Rightarrow p = 4.5 = \frac{9}{2}$.

(d) $x^2 + bx + C = 0$

$x = \frac{-b \pm \sqrt{b^2 - 4C}}{2} = 0$

$$\frac{-b-\sqrt{b^2-4c}}{2} = -\frac{1}{2}$$

$$b+\sqrt{b^2-4c} = 1 \quad -(1)$$

$$\frac{-b+\sqrt{b^2-4c}}{2} = \frac{9}{2}$$

$$-b+\sqrt{b^2-4c} = 9 \quad -(2)$$

(e) $g(2) = 2^2 + 2b + c$

$= 4 + 2b + c$

y-coordinate of vertex $= 4 + 2b + c$

(f) let $b=1$, $c=2$.

$$f(x)=g(x)$$

$$\frac{4x+2}{x-2}=x^2+x+2$$

$$4x+2=(x^2+x+2)(x-2)$$

$$4x+2=x^3-2x^2+x^2-2x+2x-4$$

$$x^3-x^2-4x-6=0$$

product of roots $=6$

11. $P(x) = 3x^3 + 5x^2 + x - 1$

(a) $P(-1) = 3(-1) + 5(1) - 1 - 1$

$= 0$

Hence, $x+1$ is a factor of $P(x)$

(b)

$$\begin{array}{r|l} & 3x^2 + 2x - 1 \\ \hline x+1 & 3x^3 + 5x^2 + x - 1 \\ & 3x^2 \\ \hline & 2x^2 + x - 1 \\ & 2x^2 + 2x \\ \hline & -x - 1 \\ & -x - 1 \\ \hline & 0 \end{array}$$

$P(x) = (x+1)(3x^2 + 2x - 1)$

$= (x+1)(3x-1)(x+1)$

$= (3x-1)(x+1)^2$

$Q(x) = (x+1)(2x+1)$

(c) $\dfrac{1}{Q(x)} = \dfrac{1}{(x+1)(2x+1)}$

$= \dfrac{1}{x+1} + \dfrac{1}{2x+1}$

(d) $\dfrac{1}{(x+1)Q(x)}$

$= \left(\dfrac{1}{x+1}\right)\left[\dfrac{1}{x+1} + \dfrac{1}{2x+1}\right]$

$= -\dfrac{1}{(x+1)^2} + \dfrac{1}{(x+1)(2x+1)}$

$= -\dfrac{1}{(x+1)^2} + \dfrac{1}{x+1} + \dfrac{1}{2x+1}$

(e) $\displaystyle\int \frac{1}{(x+1)^2(2x+1)}\,dx$

$\displaystyle = \int -\frac{1}{(x+1)^2} + \frac{1}{x+1} + \frac{1}{2x+1}\,dx$

$= (x+1)^{-1} + \ln(x+1) + \ln(2x+1)$

12. $\phi = (a+bi)^3$, $a, b \in \mathbb{R}$.

(a) $(a+bi)^3 = (a+bi)(a+bi)(a+bi)$

$= (a^2 + 2abi + b^2)(a+bi)$

$= a^3 + a^2bi + 2a^2bi + 2ab^2 + b^2a + b^3i$

$= (a^3 + 3ab^2) + (3a^2b + b^3)i$

(i) $Re(\phi) = a^3 + 3ab^2$

(ii) $Im(\phi) = 3a^2b + b^3$.

(b) $(1+\sqrt{3}i)^3$

$= [1 + 3(3)] + [3\sqrt{3} + (\sqrt{3})^3]i$

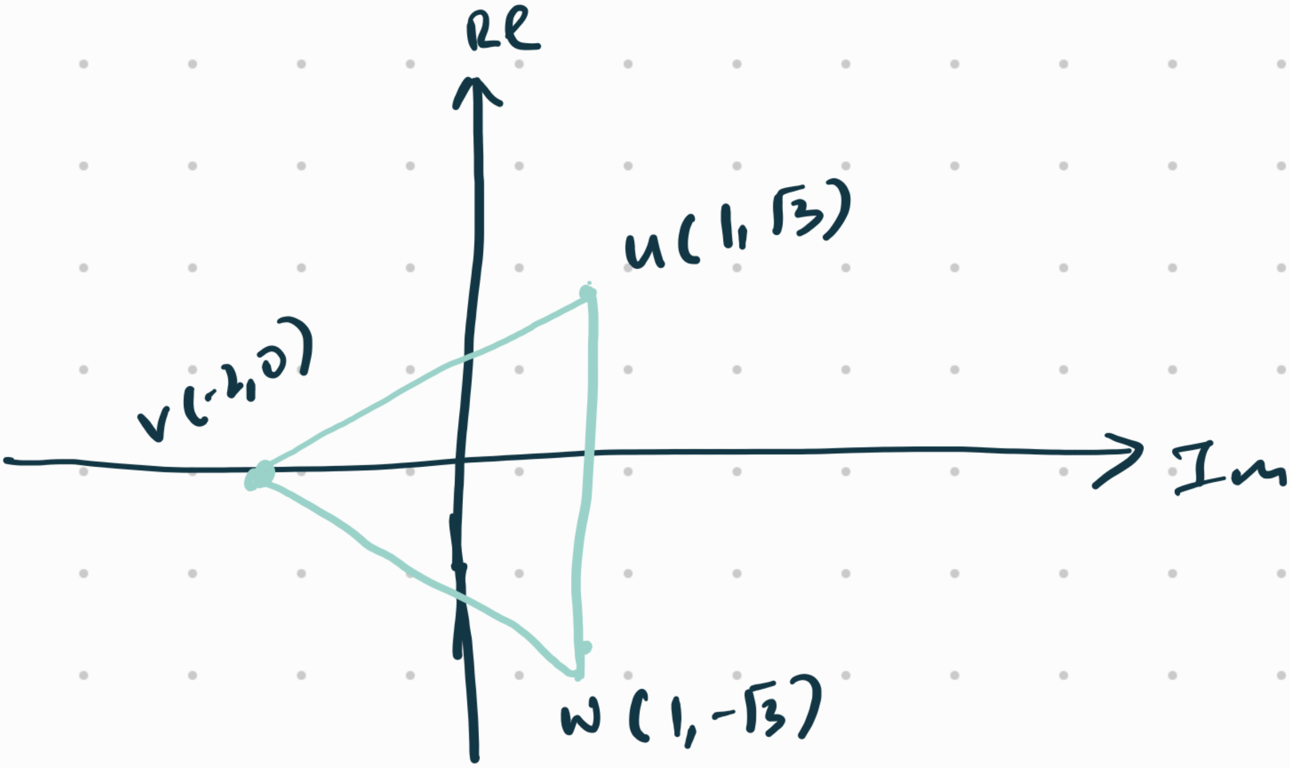
$= 10 + [3\sqrt{3} + (\sqrt{3})^3]i$

$z^3 = -8$, $\quad u = 1 + \sqrt{3}i \quad v \in \mathbb{R}$.

(c) $v = -2$.

$w = 1 - \sqrt{3}i$ (conjugate)

(d)



Area of $\triangle uvw = \frac{1}{2}(2\sqrt{3})(3)$

$= 3\sqrt{3}$ units^2

(e) $u = 1 + \sqrt{3}i = 2e^{i\left(\frac{\pi}{3}\right)}$

$v = -2 = 2e^{i\pi}$

$w = 1 - \sqrt{3}i = 2e^{i\left(\frac{5\pi}{3}\right)}$

(f) u', v', w' are solutions of $z^3 = c + di$

(g) $n = 10$